ITRI626 (2026) Update for Study Units 5, 6, 7

Introduction

The content of this module (ITRI626) is currently being revised to reflect the changing lansdcape of machine learning tools and techniques in the broad field of artificial intelligence. This transitional document reflects some of the changes that may apply to study units 5, 6 and 7.

To succeed in this module and focus on your future, your **human learning process** should include the following:

- put in "effort" to get to understand "new" things
- train your own mind to overcome obstacles while doing a task e.g. constructing a solution to a problem
- gain experience in using "thinking tools" (ranging from pen-and-paper to AI agents)
- understand the strengths and weaknesses of different types of tools, so that you choose to use them optimally to improve your own thinking and learning
- reflect on your progress and challenge yourself for each next step
- use your mind to analyse, synthesise and create new explanations ("teach it out loud")

In the past, students would get better marks if they submitted polished language and "impressive" projects / results, because that implied mastery of the techniques required to achieve that outcome. (The well-formed submissions were a proxy of the learning and understanding achieved by the student.)

However, you should not expect to "get marks for effort". You should focus your effort to build up your own understanding and ability to use the concepts and tools in solving real problems.

Learning outcomes (adapted and combined from 2025 study guide SU 5, 6, 7)

At the end of this study unit, you should:

- Understand and be able to use tools that use semi-supervised learning
- Understand and be able to use transfer learning to apply a model that has been trained in one domain to a different domain
- Understand the basic building blocks and operation of Large Language Models
- Be able to use Large Language Models effectively as they exist already
- Be able to fine-tune a Large Language Model to a specific domain

Study material

Study the handouts, slides, and video on large language models that your lecturer will distribute. This section of the course will be handled predominantly practically. Still, you are encouraged to seek as many sources as possible to contribute to your understanding beyond what will be presented in class and in the practical sessions.

Starting points:

- What Is Local AI? A Complete Guide. https://www.local-llm.net/learn/what-is-local-ai/ (Published April 8, 2026)
- Fine-tuning LLMs Guide - Learn all the basics and best practices of fine-tuning. Beginner-friendly. https://unsloth.ai/docs/get-started/fine-tuning-llms-guide (Published 2026-03-17)

Glossary

The provided spreadsheet with a .glossary for local LLMs and Agents can be used as your starting point to keep record of your own growing understanding of the following apects:

- **Model architecture and training**: Concepts that explain how models work and how they are trained.
- **Local inference and optimization**: Terms related to GPU memory, model loading, inference speed, and performance tuning.
- **Agents and tool use**: Concepts that distinguish an AI agent from a conventional chatbot.
- **RAG and knowledge management**: Terms related to document ingestion, retrieval, and long-term memory.
- **Evaluation, security, and deployment**: Concepts for measuring quality, controlling agent behavior, and operating local AI systems reliably.

You can use the glossary in different ways and include the resulting modifications or artifacts in your portfolio.

Assessment

Written Test(s)

- Based on the concepts listed in the “Glossary for local LLMs and Agents”
- Multiple Choice and short answer questions
- Case study application of the concepts

Practical assignment(s)

- Set up a system which uses a local LLM for agentic workflow(s) on the portable environment provided for this module. Suggested platform: **unsloth**
- Document your system completely, including
    - design and development
    - intro video and usage instructions
- Example topics to consider
    - A coding / SW dev agent- helper
    - Image recognition / processing workflow
    - Workflow to experimentally compare different ML approaches for a dataset and task.
    - ...
- Submission and assessment of practical work
    - During the examination period you will be asked to demonstrate your system on the portable environment to the lecturer.
    - You must also submit the portable SSD as your complete portfolio of evidence for ITRI626.
    - A one-page printed document must be submitted to show the layout and guide the examiner (and moderator) to use the portfolio efficiently and effectively.